

Divyansh Bansal

AI Engineer · LLM and voice agents, evaluation, retrieval

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Summary

AI Engineer specializing in LLM agents, evaluation, retrieval, and production data systems. Currently engineering an agentic reconciliation platform with a focus on reliability, correctness, and scalable data processing. Built a real-time voice agent and retrieval systems; published researcher with experience in Python, LangGraph, Azure OpenAI, FastAPI, PySpark, and Databricks.

Experience

AI Software Engineer | Amdocs (AT&T account), New Delhi Sep 2025 – Present

- Built, as **sole engineer**, an agentic reconciliation platform where GPT-4.1 agents handle intake, NL-to-SQL and reporting and a deterministic engine owns every verdict; LLM intake lifted completion from **71% to 96%** of 126 test requests at \$0.004 each.
- Built a reproducible evaluation platform (12+ suites, record/replay LLM proxy) that gates every merge request in CI at **\$0 API cost** and surfaced **38 ranked issues**, including the two defects below.
- Found that tables over 10k rows were silently sampled and rebuilt the read path with batched streaming, making reconciliation exact at **1M rows per side** on a synthetic benchmark (missed breaks: **97,212 → 0**).
- Fixed an agent-runtime defect that failed every run; the NL-to-SQL agent then reached **87.8% execution accuracy** on 498 benchmark questions (fixed-step pipeline: 67%) at **\$0.006** and 7.8 s p50 each.
- Designed a self-healing loop that proves each format-noise fix by re-comparison before human approval: cleared all noise on **16 of 18** test pairs while masking **0 real breaks** across 17,842 out-of-sample cells.
- Cut a production stored-procedure run over **75M records/day** from **45 to 12–15 min**; leading the migration (about 80% complete) of the subscriber-event pipeline to Azure Databricks (PySpark, Unity Catalog).
- Led technical engagements with **40+ Amdocs and AT&T stakeholders** through technical discussions, solution demos, and POCs, translating feedback into product iterations.

Co-Founder & Product Architect | DTU ERP Portal, New Delhi | dtu-eceportal.in Oct 2024 – Jul 2025

- Co-founded and architected an ML-powered ERP platform for **1,200+** active university users; semantic search (embeddings, ChromaDB) cut accreditation prep from **25 to 7 days**.
- Trained pilot-department faculty and staff, supported the rollout, and pitched the platform to the university administration, which **now funds it**.

Projects

VocalisAI: Voice Agent That Calls Airlines for Refunds | Open source [Demo](#) | [Code](#)

- Built a real-time voice agent that negotiates airline refunds under Indian, UK/EU and UAE passenger-rights rules (Pipecat, LangGraph, Deepgram), with **91% task success** over 75 adversarial simulated calls in the text harness.
- Kept secrets out of the model's context (placeholders resolved only at speaking time) behind a deterministic streaming output guard: **0 sensitive-data leaks** across 75 adversarial calls.
- Recalibrated a ceiling-biased LLM judge (29/30 calls rated 5/5) with an anchored rubric, raising agreement with an independent LLM labeller from κ **-0.05 to 0.73** on 30 held-out calls.
- Built jurisdiction-scoped hybrid retrieval (BM25, dense embeddings, cross-encoder reranking) for regulation citations, lifting top-1 accuracy from **47% to 65%** on 78 gold questions.
- Measured **1.86 s p50 / 2.70 s p95** voice-to-voice latency on a simulated 8 kHz phone line; a phonetic read-back fix lifted task success over audio from **56% to 68%** (25 calls each).

IntelliCode: Hybrid RAG and Code Analysis | Open source [Demo](#) | [Code](#)

- Built hybrid RAG (BM25 + FAISS, RRF, cross-encoder reranking) serving Qwen2.5-3B on Hugging Face ZeroGPU, plus an AST analyser for 12 anti-pattern classes; **121 tests at 92% coverage**.

Technical Skills

LLMs & Agents: Azure OpenAI, open models (Qwen2.5), LangGraph, LangChain, MCP, LiteLLM, tool calling, structured outputs, prompt and context engineering, guardrails, human-in-the-loop, Pipecat, Deepgram (streaming STT/TTS), simulation testing

Retrieval & Evaluation: RAG, hybrid search (BM25, dense, reranking), FAISS, ChromaDB, LLM-as-judge, Cohen's κ , ablations, CI regression suites, cost/latency optimisation

Backend & Data: Python, SQL, FastAPI, Celery, PostgreSQL, Oracle, Redis, PySpark, Azure Databricks, Unity Catalog, Airflow

ML, Infra & Frontend: PyTorch, scikit-learn, Docker, GitHub Actions, GitLab CI, Google Cloud Run, React

Publication

“Design and Implementation of a Multiuser Enterprise Resource Planning Solution for Higher Education Institutions: Enhancing Accreditation and Administration.” *Software: Practice and Experience* (Wiley), 2026 | DOI: [10.1002/spe.70060](https://doi.org/10.1002/spe.70060)

Education & Certifications

B.Tech, Electronics and Communication Engineering, Delhi Technological University (DTU), 2021 – 2025 | CGPA 8.06 / 10

Databricks Certified Data Engineer Associate | Databricks Certified Generative AI Engineer Associate